

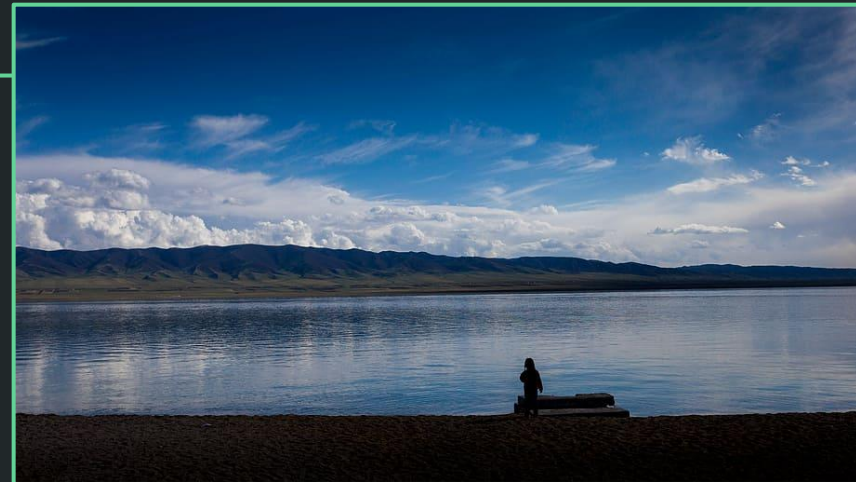


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Qinghai Lake's Surface Area Change

Remote Sensing & Climate Change

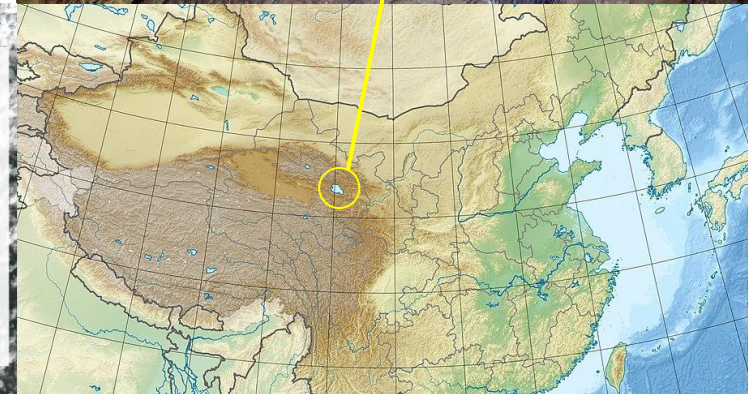
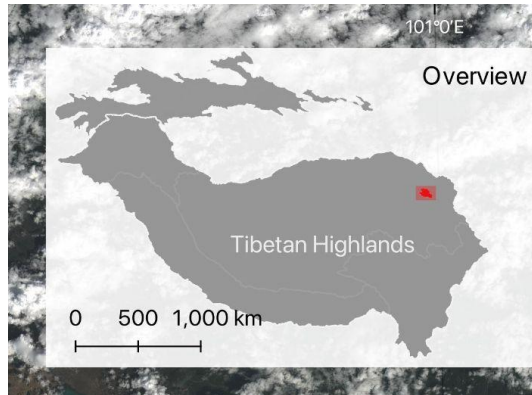
Sentao ZENG
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University of Milan | June 2023



1. Introduction

Qinghai Lake:

- The largest lake in China (4,529.5 Km², 2022)
- Located in an endorheic basin in Qinghai Province, China.
- Lies in a hollow of the **Tibetan Plateau** at 3,205 metres above sea level.
- Classified as a saline and alkaline lake.
- There's a mountain in the middle of the lake called Haixin Mountain, as well as a few other small islands.



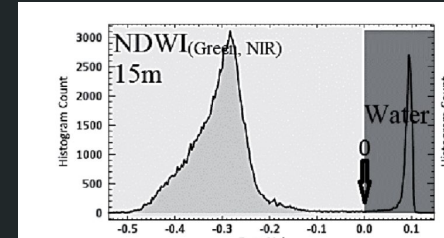
2. Method

Analyze remote sensing data to study the change of the surface area of Qinghai Lake.

1. Satellite images source: USGS; Landsat 7 for 2002, Landsat 9 for 2022.
 2. Use QGIS to calculate the Normalized Difference Water Index (NDWI)
- $$NDWI = \frac{Green - NIR}{Green + NIR}$$
3. Choose a proper threshold value to distinguish water surface.
 4. Polygonize the water surface area of the lake and get the relevant statistics values.
 5. Compare the results between 2002 and 2022 and draw a conclusion.

NDWI threshold for lake water:

- 0 (2022, Landsat 8-9)
- 0.1 (2002, Landsat 7 ETM+)



Emre Özelkan, 2019

TABLE 2. NDWI and NDVI THRESHOLDS FOR PURE WATER ($f_w = 100\%$) AND MIXED WATER ($f_w = 25\%$, 50% , AND 75%) COMPONENTS

Sensor		NDWI and NDVI threshold									
		$f_w = 25\%$					$f_w = 100\%$				
		$f_s = 75\%$	$f_s = 37.5\%$	$f_s = 0\%$	$f_s = 50\%$	$f_s = 25\%$	$f_s = 0\%$	$f_s = 25\%$	$f_s = 12.5\%$	$f_s = 0\%$	$f_s = 0\%$
Landsat ETM+	NDWI ₁₂₄	-0.351	-0.553	-0.657	-0.313	-0.512	-0.619	-0.236	-0.416	-0.528	0.015
	NDWI ₂₄₂	-0.611	-0.563	-0.504	-0.569	-0.519	-0.459	-0.464	-0.414	-0.356	0.123
	NDWI ₁₆₇	-0.587	-0.458	-0.223	-0.542	-0.410	-0.181	-0.433	-0.300	-0.096	0.179
SPOT-5	NDWI ₁₆₇	0.223	0.569	0.782	0.201	0.527	0.735	0.156	0.430	0.622	-0.007
	NDWI ₆₁₃	-0.394	-0.595	-0.672	-0.350	-0.535	-0.633	-0.261	-0.433	-0.539	0.016

Vivek Kumar Gautam et al, 2015



NDWI Threshold = 0.2



NDWI Threshold = 0.1

Sentao Zeng, 2023

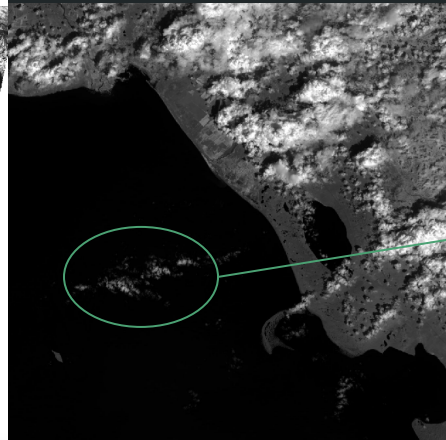
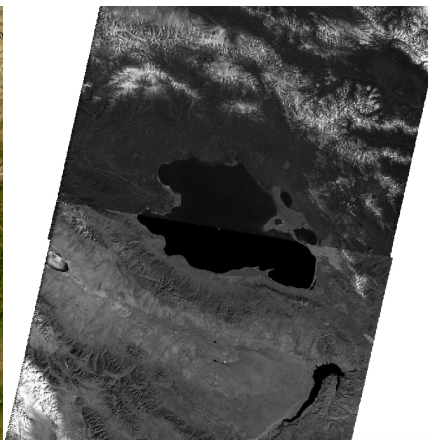
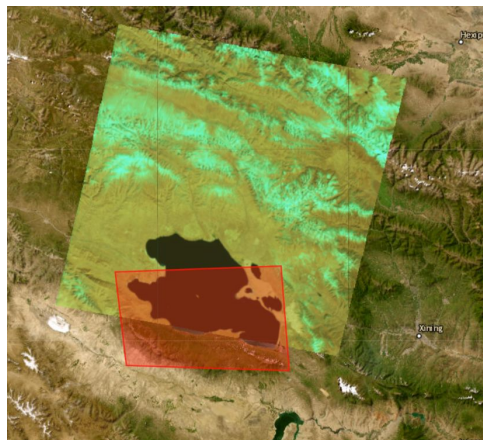
3. Satellite Imagery

1. The lake is “cut”

There's not a single satellite image that covers the whole lake between the year of 1998-2020.

Solution:

Merge 2 satellite images with sequential path-row numbers. (e.g. *LE07_L1TP_133034* & *133035*) Prepare such new sets of images for each band before processing.



2. Ice & clouds issues

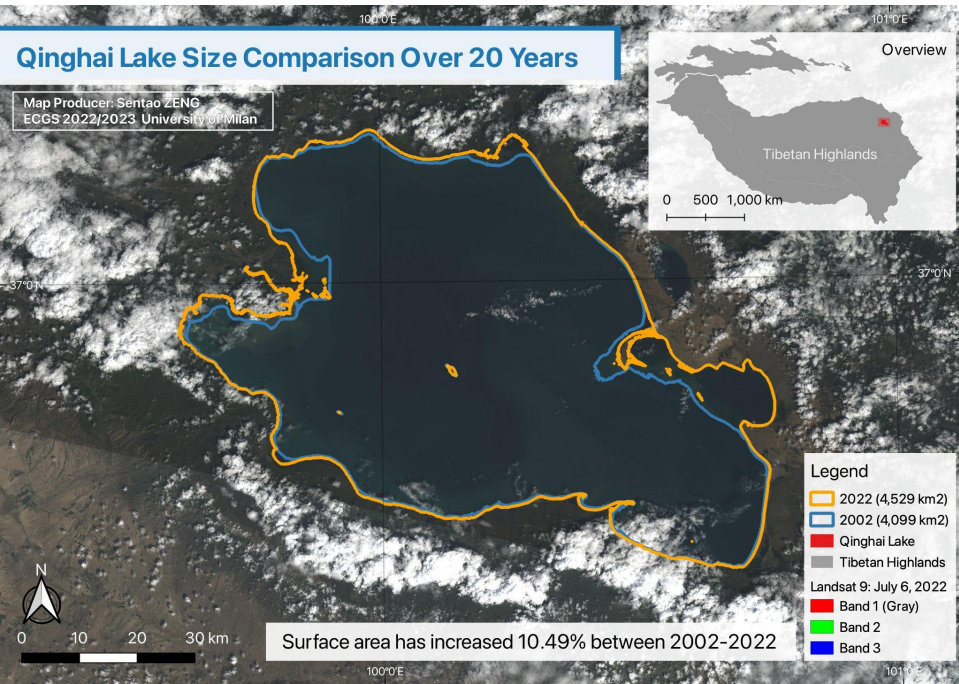
In the first attempt of making the map, the satellite images looked fine with human eye. But after NDWI calculation, there's difficulty distinguishing water surface due to clouds and ice.

Solution:

With careful selection on USGS Earth Explorer, image taken in July 2022 and November 2002 were used. (Qinghai lake usually starts to freeze from December to March)

4.Results

Year:	Area of Interest (m^2)	Area of Interest (km^2)
2002	4.10E+09	4,099.40
2022	4.53E+09	4,529.53



Using QGIS Statistics Tool to Calculate the Surface Area of Qinghai Lake

Statistics

year22-area-no-cloud

\$area

Statistic	Value
Count	1
Sum	4.52953e+09
Mean	4.52953e+09
Median	4.52953e+09

✓ Selected features only

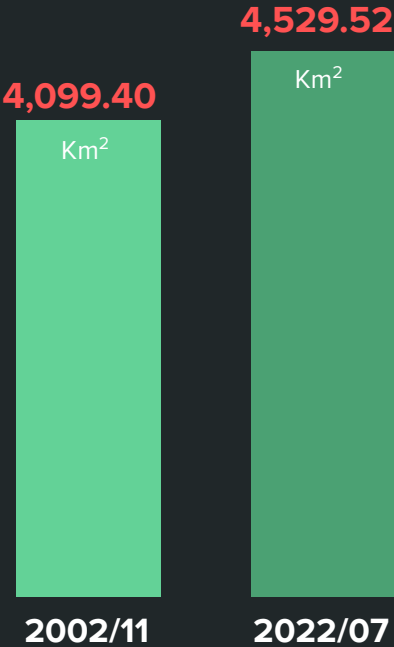
Statistics

year02-area

\$area

Statistic	Value
Count	1
Sum	4.0994e+09
Mean	4.0994e+09
Median	4.0994e+09

✓ Selected features only



Expansion:
10.49% ▲

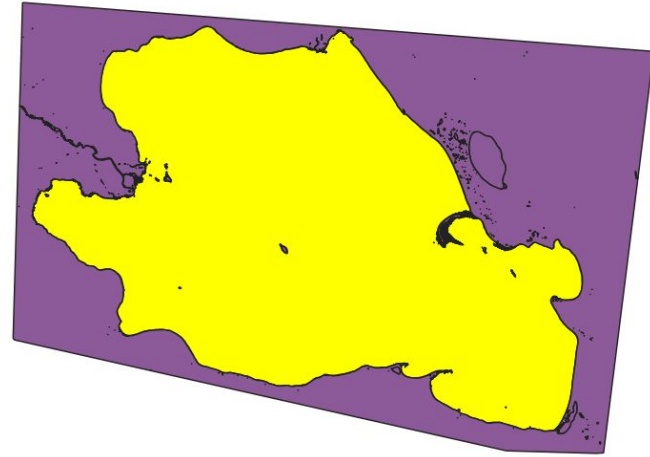


Discussion

The expansion of Qinghai Lake over the last 20 years can be mainly attributed to:

- The rapid increasing annual precipitation after 2004;
- The decline in daily temperature range and thus decreasing evaporation.

(Hongmei Dong et al, 2022)



Thank You!

References:

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2. Vivek Kumar Gautam, Piyush Kumar Gaurav, P Murugan, M Annadurai. Assessment of Surface Water Dynamics in Bangalore using WRI, NDWI, MNDWI, Supervised Classification and K-T Transformation. *International Conference On Water Resources, Coastal And Ocean Engineering*, (2015)
3. Hongmei Dong, Yougui Song, Maosheng Zhang. Hydrological trend of Qinghai Lake over the last 60 years: driven by climate variations or human activities? *Remote Sensing*, (2022)
4. Hongyu Duan, Xiaojun Yao, Dahong Zhang, Huian Jin, and Qixin Wei. Long-Term Temporal and Spatial Monitoring of Cladophora Blooms in Qinghai Lake Based on Multi-Source Remote Sensing Images. *Remote Sensing*, (2022)